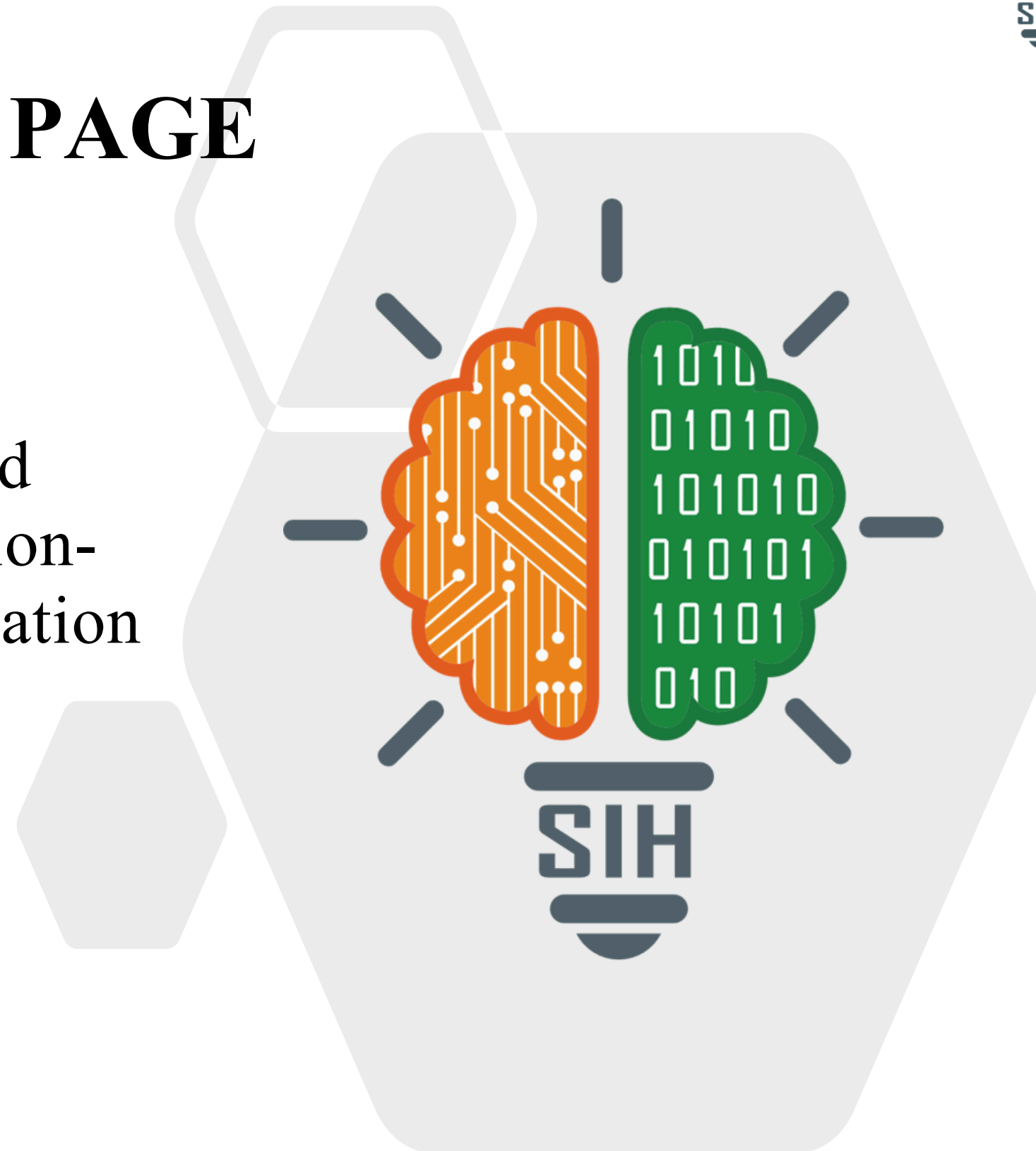
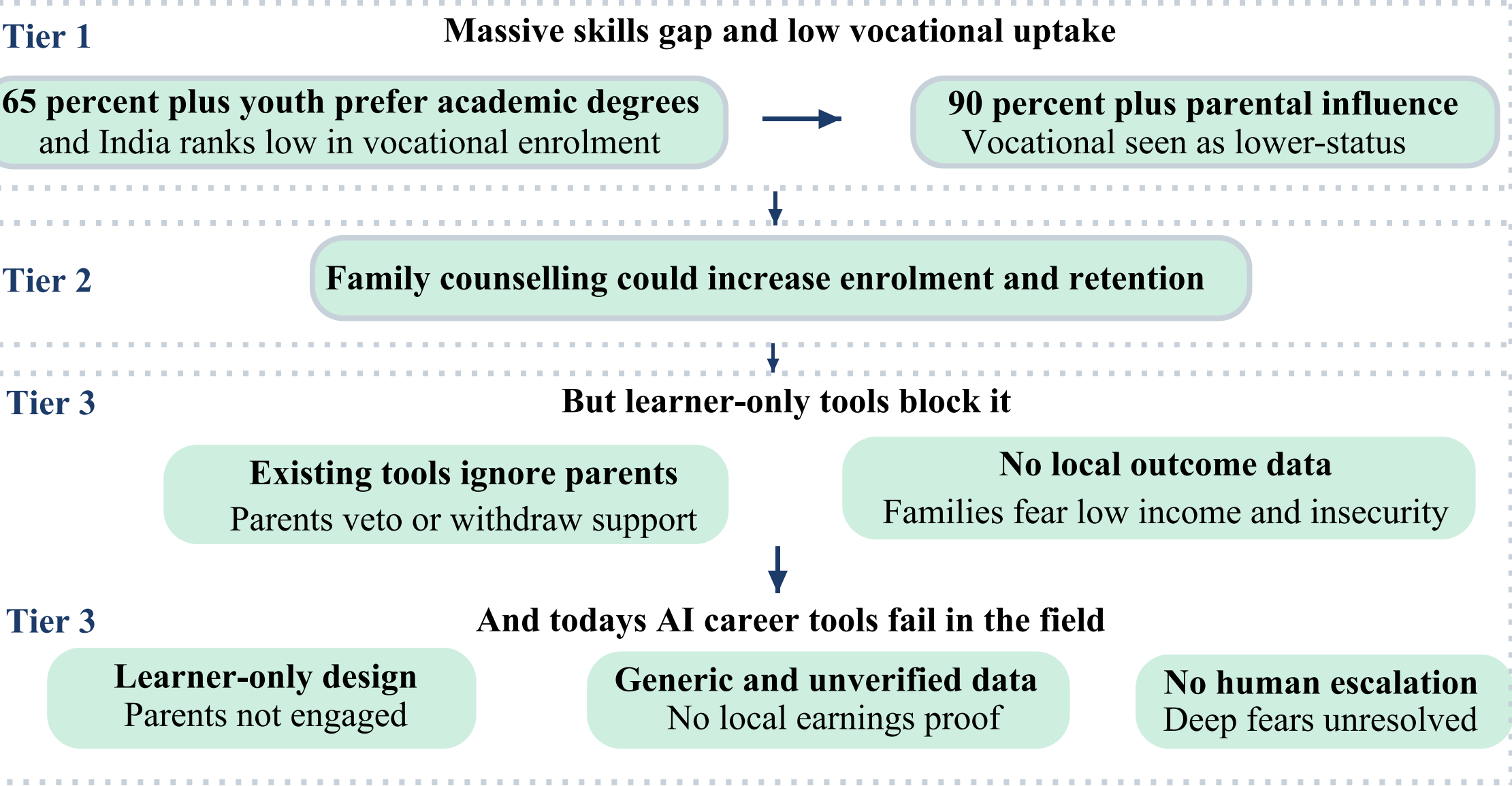


TITLE PAGE

- **Problem Statement ID** – SIH26241
- **Problem Statement Title-** AI-Enabled Career Counselling and Family Decision-Support Platform for Vocational Education
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 173175
- **Team Name:** TheBIG(O)



The Problem



Proposed Solution

Voice-First Dyadic PWA: A Bhashini-powered conversational PWA across 22 regional languages that counsels learners and parents simultaneously using dedicated "Learner" and "Parent" modes.

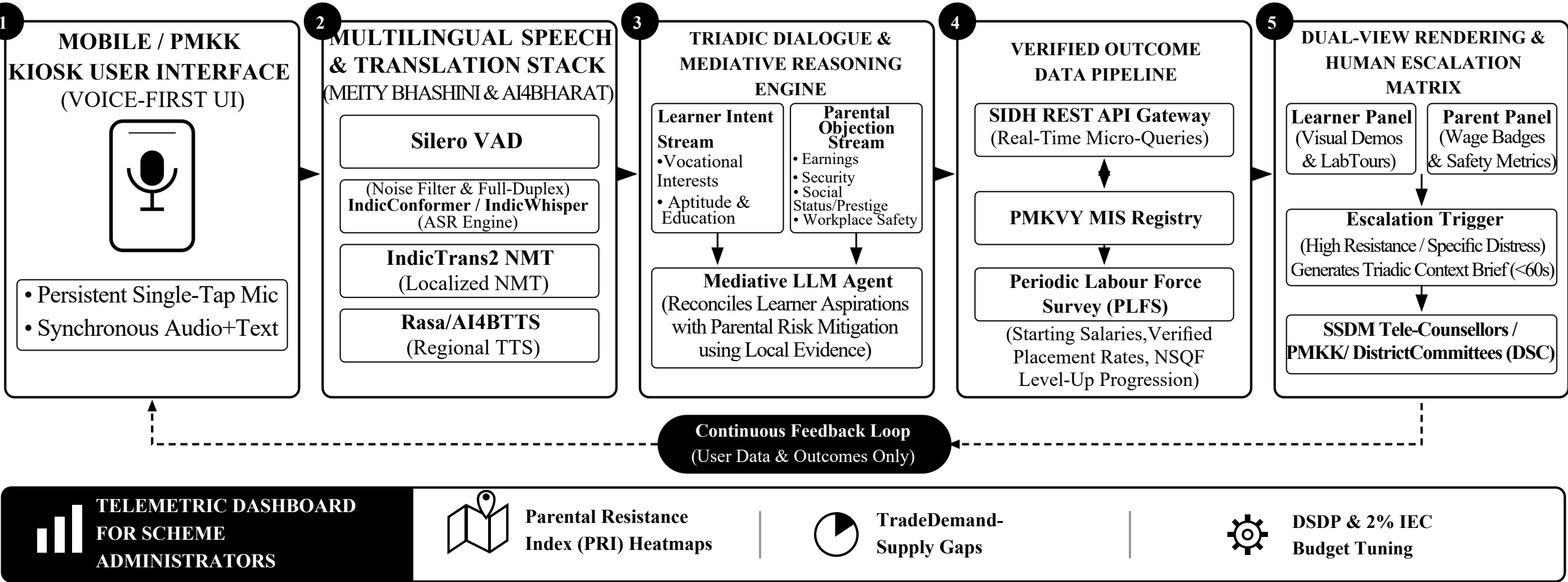
Verified DPI Middleware: A unified backend pulling real-time district placement wages (SIDH JobX), stipends (NAPS), and credit equivalencies (NCrF / ABC).

Innovation & Uniqueness

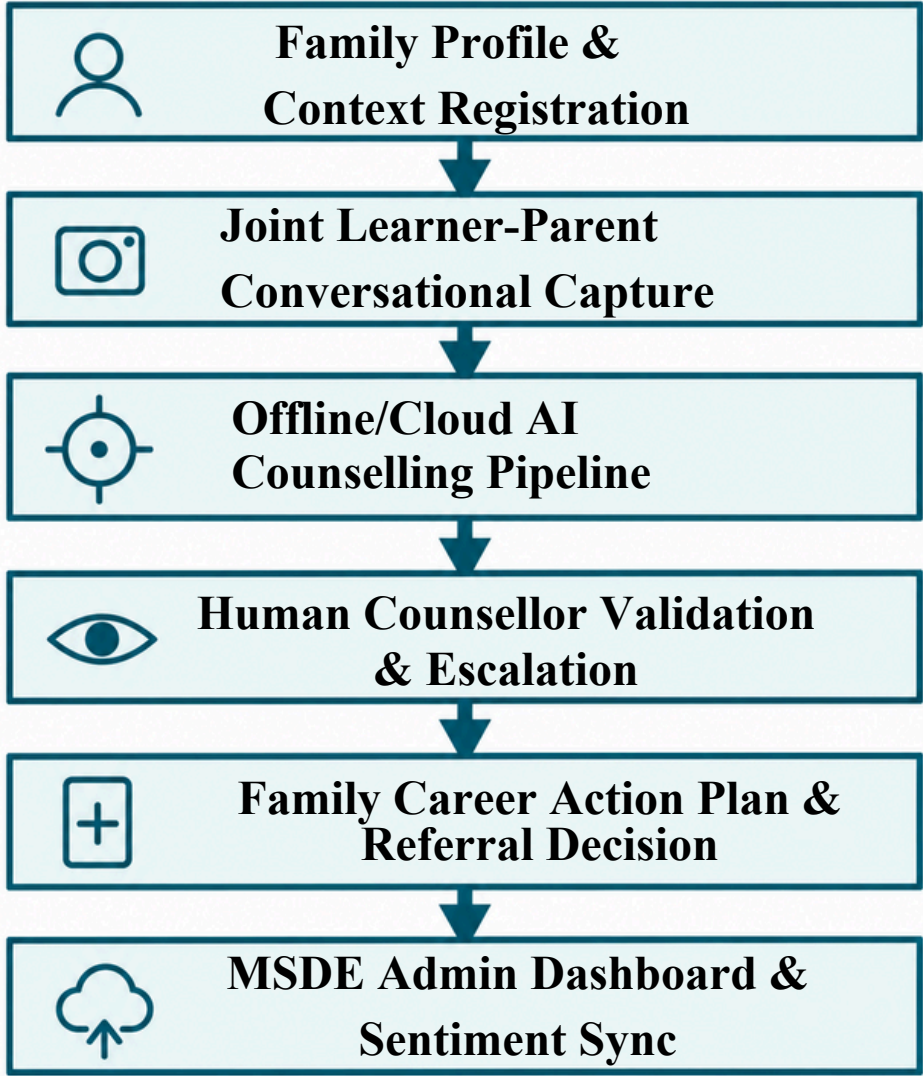
Dual-Speaker Family Decision Engine: The first guidance system built for collective family consensus rather than individual course search.

Algorithmic Escalation (Rs > 0.75): Computes a real-time Parental Resistance Score (Rs > 0.75) to trigger instant WebRTC handoff to live PMKK counsellors

How It Addresses the Problem		
Data-Backed Reframing: Replaces social stigma with verified local placement data, guaranteed NAPS stipends, and center safety ratings (female ratios, CCTV, transport).	Guaranteed Degree Mobility: Deposits NSQF credits into the student's Academic Bank of Credits (ABC / APAAR ID) to preserve lateral entry into university degrees (B.Voc/B.Tech).	Targeted Governance: Aggregates anonymised telemetry into district-level GIS heatmaps so MSDE/NSDC administrators can deploy targeted ground interventions where resistance is highest.



KaushalSakalpAI: End-to-End App Flow



Technical Architecture & Stack

Dual-Speaker Family Interface
Technology: React PWA + Voice/Text UI
Tech: Learner Mode + Parent Mode with separate aspiration and concern inputs.
Impact: Enables joint family counselling through a low-literacy, multilingual and voice-first experience.

Indic Voice Intelligence
Technology: BHASHINI / Dhruva Language Stack
Tech: ASR + NMT + TTS for regional-language speech interaction.
Impact: Converts vernacular voice into structured intent and returns simple audio-visual guidance.

Contextual AI & Resistance Engine
Technology: RAG + LLM + FastAPI
Tech: Intent detection, concern classification, family-context reasoning and proposed RsR_s scoring.
Impact: Personalises counselling, addresses parental concerns and identifies high-friction cases for escalation.

Verified Outcome Data Middleware
Technology: PostgreSQL + pgvector + API Middleware
Tech: Trade, provider, location, earnings, employment, apprenticeship and progression evidence with source metadata.
Impact: Replaces generic career claims with evidence-linked, locally relevant information.

Feasibility and Viability

Technical Feasibility

Evidence-First AI
RAG + Verified Outcome Data:
Grounds LLM responses in source-linked career, earnings, placement and progression evidence, with metadata and claim validation to reduce unsupported AI-generated information.

Hybrid Cloud-Edge
Low-Bandwidth Architecture:
Lightweight PWA, cached content and asynchronous sync support constrained rural connectivity.

Financial Feasibility

Low Infrastructure Cost
Existing Digital Infrastructure:
Reuses BHASHINI, Skill India Digital, NCS, NCVET/NCrF and ABC/APAAR ecosystems.

Zero User Cost
Accessible Deployment: Free for learners and families through smartphones, CSC/ kiosk devices and lightweight web infrastructure.

SKILL ECOSYSTEM READINESS

- 1.64 crore+ youth trained under PMKVY
- 1.5 crore+ candidates registered on Skill India Digital Hub
- 56.08 lakh+ apprentices engaged since 2016

Operational Feasibility

Minimal Training Requirement
Voice-First Interaction:
Regional-language voice, visual cards and guided workflows require minimal digital orientation.

AI + Human Escalation
Counsellor Case Pack: AI handles routine queries and prepares high-friction cases for human counsellor intervention.

Social Feasibility

Family-Centred Counselling
Learner + Parent Modes:
Captures aspirations and concerns around income, security, safety, status and career growth.

Inclusive Career Access
Regional & Low-Literacy Support: Multilingual voice guidance and simple pathways improve access for underserved households.

Strategies for Overcoming Challenges

Technical Challenge

Challenge: AI may generate generic or unsupported career, salary and placement claims.
Mitigation: RAG + verified outcome data + claim validation grounds responses in source-linked evidence.

Financial Challenge

Challenge: Multilingual AI and national-scale deployment can increase infrastructure and maintenance costs.
Mitigation: Reuses BHASHINI, Skill India Digital, NCS and existing DPI with a lightweight PWA to reduce deployment overhead.

Operational Challenge

Challenge: Complex family concerns require human judgement while counsellors face repetitive queries.
Mitigation: AI handles routine guidance; high-friction cases ($R_s > 0.75$) trigger human counsellor escalation with a structured case pack.

Target Customers & Stakeholders

1. Institutional Buyers & Policy Makers:

State Skill Development Missions (SSDMs), MSDE, and District Skill Committees managing public skilling budgets, DSDPs, and IEC funds.

2. Operational End-Users:

Frontline tele-counsellors, PMKK center coordinators, District Skill Officers, and community mobilization teams.

3. Primary Beneficiaries

Rural and semi-urban learners alongside their parents, making joint, risk-mitigated career decisions.

Offline & Kiosk Continuity

Enables uninterrupted rural voice guidance.



Scalable Registry Integration

Synchronizes live with SIDH databases



Rapid Human Escalation

Connects live counsellors within seconds



Dropout Stigma Reduction

Prevents mid-course dropouts before enrolment



Impact on Target Audiences



Requires zero textual reading skills

Low-Literacy Voice Interface



Replaces hearsay with real wages

Verified Outcome Data



Shields families from graduate unemployment

Degree Trap Protection

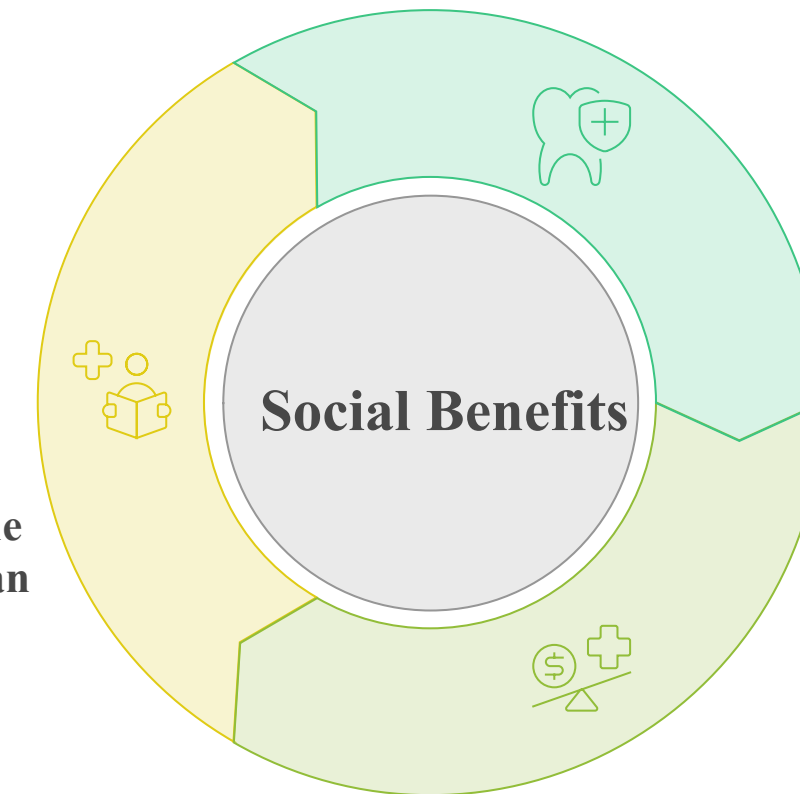


Reaches underserved rural NEET youth

Rural Youth Expansion

Destigmatize Vocational Trades

Reframes ITI and PMKVY courses into aspirational, high-value career paths rather than second-tier fallbacks



Multilingual Family Inclusion

Uses Bhashini speech AI so low-literacy parents engage naturally in native dialects while addressing female workplace safety.

Transparent Outcome Data

Replaces informal hearsay with verified starting wages, placement rates, and NSQF progression paths directly from Skill India Digital Hub (SIDH).

Economic & Operational Benefits

- 1. Mitigates Educated Unemployment:** Diverts youth from general degrees (13% overall / 20.4% female unemployment) to NSQF vocational tracks with verified starting wages and >60% placement.
- 2. Direct Family Savings:** Delivers voice guidance on mobile/PMKK kiosks, eliminating private counselling and travel expenses for rural households.
- 3. Amplifies Counsellor Capacity:** Automated briefs and objection classification boost SSDM tele-counsellor operational efficiency by 5x.
- 4. Targeted IEC Funding:** Helps District Committees allocate 2% awareness budgets directly to specific resistance drivers identified by the Parental Resistance Index (PRI).

References

- [1] NITI Aayog (2026) — Reimagining Skilling for Viksit Bharat@2047 — Pathway Clinics, MyGuide, Skill Scout and National Career Outcome Repository.
- [2] MoSPI — PLFS 2023–24 — vocational and technical training participation among youth.
- [3] [MSDE / PIB (2026)] Skills, Scale and Transformation — PMKVY, Skill India Digital Hub & Apprenticeship Ecosystem.
- [4] [MSDE / PIB (2026)] Digital Skill Development Platform — Skill India Digital Hub (SIDH).
- [5] NCVET / NCrf — National framework for vocational qualification levels, credits and progression.
- [6] BHASHINI / Digital India Corporation — Multilingual AI, speech and language services for Indian languages.
- [7] Research on Indian VET Perceptions — Studies examining parental and societal perceptions of ITIs and polytechnics.
- [8] Schneider (2023/2024)
The Attractiveness of Polytechnics in Delhi and Mumbai — Student & Parent Perceptions.
- [9] Ajithkumar, U. & Pilz, M. (2019)
Attractiveness of Industrial Training Institutes (ITI) in India: A study on ITI students and their parents. Education + Training, 61(2), 153–168.

Live Demo:
GitHub Repo:
Prototype Video: